

Koby Lewis

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Education

- BlueDot Technical AI Safety Course
- Brigham Young University, PhD of Computer Science, Anticipated Graduation Apr 2028
 - Prosocial Language Models Emphasis
- Brigham Young University, MS of Computer Science, Graduated Aug 2025
 - Programming Languages/Software Engineering Emphasis
- Brigham Young University, BS of Computer Science, Graduated Apr 2023
 - Valedictorian, Summa Cum Laude, GPA 4.0
 - Software Engineering Emphasis
 - Linguistics Minor

Work/Research Experience

- Brigham Young University, Aug 2024-Present
 - Research Assistant under Dr. Nancy Fulda (Machine Learning), Aug 2025-Present
 - Organizing and leading development on Large Language Model training pipeline
 - Research Assistant under Dr. Kimball Germane (Programming Languages), Aug 2024-Aug 2025
 - Implemented FlowDB, an implementation of demand-driven control-flow analysis for TypeScript.
 - Produced and defended a Master's Thesis in one year
- Awardco, Apr 2022 – Aug 2024
 - Backend Software Engineer, Dec 2022 – Aug 2024
 - Accelerated database update speed by over 200x across the platform
 - Architected and built Queue and Delta based user update system
 - Decoupled build and development process from relying on a single IDE
 - Backend Software Engineering Intern, Apr 2022 – Dec 2022
 - Expanded flexible system for copying client settings
 - Implemented roles and permissions system
- Brigham Young University IDeA Labs, Fall 2022
 - Undergraduate Researcher
 - Organized a team of undergraduate researchers to produce a conference publication in Control Theory
- Lucid Software, May 2021 – Aug 2021
 - Software Engineering Intern

- Developed pages for onboarding surveys to drive engagement and virality
- Brigham Young University Center for Teaching and Learning, Fall 2019 – Apr 2021 and Aug 2021 – Apr 2022
 - Web Developer, Mar 2020 – Apr 2021 and Aug 2021 – Apr 2022
 - Implemented various pages for the University LMS (Learning Management System)
 - Created online discussion forum that allowed users to record and post videos of themselves
 - Heavily refactored and extended exam proctoring system
 - Quality Assurance Fall 2019 – Mar 2020
 - Thoroughly searched for, found, and reported bugs in the University LMS

Skills

- Programming Languages/Frameworks: JavaScript/TypeScript, C#, Python, PyTorch, SQL, Vue, Angular, Tailwind, HTML, CSS, Java, Scala, C/C++, PHP, GLSL, HLSL, CUDA
- Technologies: AWS, Azure, Git, Bash, SQL Server, Unity

Honors and Awards

- Brigham Young University (Undergraduate)
 - Valedictorian, Summa Cum Laude
 - Full Ride Academic Scholarship
 - Dean's List every semester - College of Physical and Mathematical Sciences
- Awardco
 - Product Hackathon 2023 Winner
 - Platform Hackathon 2023 Winner
- “Outstanding Engineering” Award - Lucid Hackathon 2021

Certifications

- Language Certificate - Spanish Fluent - Brigham Young University

Projects

- [LLM training pipeline for unifying efforts across several research thrusts \(Pytorch, Lightning\)](#)
- [Experiments finding helical representation of turn taking in an LLM](#)
- [Experiments refuting “correct answer feature” hypothesis for LLM MCQA](#)
- [Port AI of Ethics benchmark to Inspect](#)
- Experiments refuting connection between ROME and Planning Circuits
- [SAE implementation on toy convolutional model \(Pytorch, Lightning\)](#)
- Transformer language model, implemented from scratch and trained (Python, Pytorch tensors)
- [Static analyzer for finding all database queries that could be executed by a function \(TypeScript\)](#)
- [Natural-language-like toy programming language interpreter \(F#\)](#)

- High-performance GPU-accelerated grid-simulation system, with code-base shared between CPU and GPU (Rust)
- App for communication and coordination during disasters (Swift, Flutter)
- Framework for quickly prototyping card and board games online (TypeScript, Vue, Express)
- Chrome extension for navigating the web with only a keyboard (JavaScript)
- Biological evolution simulator (C++, OpenGL, Neural Networks, Model-View-Presenter, Multithreading, Time Travel)
- Discord bot for managing 2000 person server (TypeScript)
- Roguelike chess-based video game (C#, Unity)
- Randomized phonemic inventory generator for constructed languages (Java)
- Multiplayer online card games (Vue, JavaScript, Express)

Volunteer Work

- BYU Computer Science Graduate Student Society, Aug 2024-Present
 - Vice President of Activities: organized weekly+ activities for graduate students to connect with one another
- BYU Computer Science Discord Server, Dec 2021 – Apr 2023
 - Head Administrator
 - Continued to increase popularity after the graduation of the founders of the group
 - Planned and coordinated in person social meetups
 - The only organization not officially tied to the university to be included during a club rush event
- The Church of Jesus Christ of Latter-day Saints, Aug 2017 – Jul 2019
 - Volunteer Missionary
 - Taught immigrants English and life skills
 - Converted public data into effective techniques for finding people to help

Publications and Presentations

- *Research Directions* – Symposium on Faith & Ethics in AI, April 2026
- *FlowDB: A Complete Method for Extracting Database Queries from Functional Programs* – Master's Thesis, July 2025
- *Development of a Mutable Meme Model in the Context of a Contagion Spread Simulation* – IEEE CCTA 2023
- *The Hero's Journey and the Plan of Salvation* – Brigham Young University Religious Education Symposium 2019